Unit 3 (Part 6) – Factorisation

Factorising is the process of rewriting many terms as one term (i.e. a product of factors). It can be seen as "undoing" multiplication.

In Grade 9, you learnt three methods of factorisation (viz. removing a highest common factor; difference of two squares; simple quadratic trinomials), and **these always should be applied in this order**:

1. Highest Common Factor (HCF)
   - Grouping … Gr. 10

2. Difference of **two** squares (DOTS)
   - **Two** cubes… Gr. 10

3. **TRI**nomial methods
   - The Factor Theorem… FSM & Gr. 12

❶ Highest Common Factors

Factorisation by removing a (highest) common factor 'undoes' the simple version of the Distributive Law.

For each example, fully factorise the expression by removing a common factor.

Eg.1) $15a^2b^3 - 6a^2b^2c + 81a^3b^2c - 102a^2b^2c$

Eg.2) $-40x^2 - 5x + 20$

Eg.3) $x(x - y) + y(x - y)$

Eg.4) $2(p - 1) + (1 - p)(p + 2) - 2(p - 1)^2$

Eg.5) $ax + ay - bx - by$

Homework: Ex. 6 pg. 19 – 20 (Mind Action Series new edition, 2022)

❷ Grouping

Often times, expressions containing many terms are unable to be factorised as a whole, and require a pre-factorisation before applying one of the three main factorising techniques.
***Grouping enables us to factorise further*.**

Fully factorise each expression given below.

Eg.1) $4x^2 + 20x - 3xy - 15y$

Eg.2) $a(2 - x) + b(x - 2)$

Eg.3) $x^3 - 5x^2 + 3x - 15$

Eg.4) $-3x^3 + 6x^2 - 15x + 30$

Homework: Ex. 11, question A pg. 30, (Mind Action Series new edition, 2022)

❸ Difference of Two Squares: **Fully** factorise each expression.

Eg.1) $p^2 - 25$

*Eg.2) $X^2 - 3$

Eg.3) $x^8 - 1$

Eg.4) $32 - 2t^2$

*Eg.5) $\frac{3p}{x^2} - \frac{27y^6}{p}$

Eg.6) $(a+1)^2 - \frac{1}{4}$

Eg.7) $1 - 900(a+1)^2$

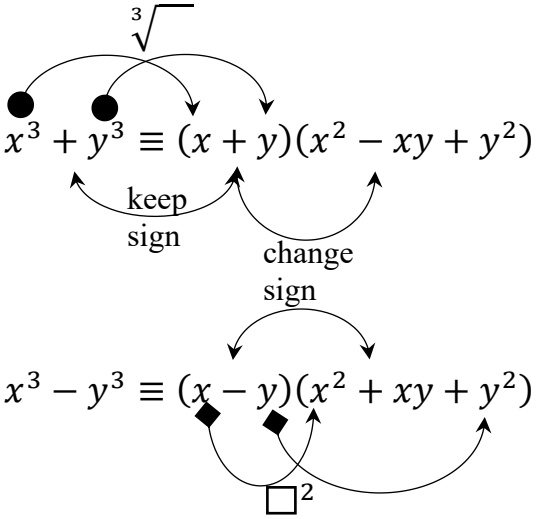
Eg.8) $(2x+y)^2 - (x-2y)^2$

Eg.9) $x^2 - 3x + 3y - y^2$

Homework: Ex. 7 pg. 21 – 22 (Mind Action Series new edition, 2022)

❹ Two Cubes

Any expression of the form $(A)^3 \pm (B)^3$ is able to be factorised into a product of a linear factor and a quadratic factor. The quadratic factor of these perfect cubes always has no real solutions so don't try to factorise it further.



Step 1: ●Cube-root each term to create the linear factor and **keep the sign**.

Step 2: ◆Working from the linear factor, square each term to create the first and last terms of the quadratic factor – remember that *anything squared is positive*.

Step 3: To create the middle term of the quadratic, take the product of the terms in the linear factor and **change the sign**.

Eg.1) Factorise $27a^3 - 64b^6$

Eg.2) Fully factorise $xy^3 + x$

*Eg.3) Fully factorise $(x^6 - y^6)$ in <u>two different ways</u>, and hence prove that $(x^4 + x^2y^2 + y^4)$ can be factorised as $(x^2 + xy + y^2)(x^2 - xy + y^2)$. How do we know for certain that none of these trinomials have any real roots?

<u>Homework:</u> Ex. 8 pg. 23 (Mind Action Series new edition, 2022)

❺ Trinomials: $ax^2 + bx + c$

- **Simple Trinomials (where $a = 1$)**

When factorising trinomials one tries to find factors of the constant, c, that sum to b, the co-efficient of x, that is the 'middle term'.

$x^2 + bx + c = x^2 + px + qx + pq$; where $b = p + q$ and $c = pq$
$= (x + p)(x + q)$

Fully factorise the following expressions.

Eg.1) $a^2 + 7a + 12$

Eg.2) $a^2 - 7a + 12$

Eg.3) $a^2 + a - 12$

Eg.4) $a^2 - 7a - 12$

Eg.5) $x^2 - 7x - 60$

Eg.6) $x^2 - 5x + 6$

Eg.7) $x^2 - 5x - 6$

Eg.8) $3a^2 - 21a - 24$

Eg.9) $-x^2 + 8x - 12$

Eg.10) $x^3 - 10x^2 + 24x$

- **Perfect Square Trinomials: (where *a* = 1)**

Perfect square trinomials have a **repeated factor**, and always produce values that are perfect squares for all rational values of *x* and *y*:

$$(x+y)^2 = (x+y)(x+y) = x^2 + 2xy + y^2$$
$$(x-y)^2 = (x-y)(x-y) = x^2 - 2xy + y^2$$

Factorise each example:

Eg.11) $x^2 - 2xy + y^2$

Eg.12) $x^2 + 6x + 9$

Eg.13) $4x^4 - 12x^2y + 9y^2$

Eg.14) $12a^4x + 36a^2b^3x + 27b^6x$

Homework: Ex. 9 pg. 25 (Mind Action Series new edition, 2022)

- **More General Trinomials: (where $a \neq 1$)**

For these types of trinomials the factors of a also have to be considered. We are 'undoing' grouping…

$$\begin{aligned}(sx + p)(tx + q) &= (st)x^2 + (sq)x + (pt)x + pq \\ &= (st)x^2 + (sq + pt)x + pq \\ &\equiv ax^2 + bx + c \qquad ; \text{ where } a = st,\ b = sq + pt,\ c = pq\end{aligned}$$

Fully factorise each example.

Eg.1) $21x^2 + 25x - 4$

Eg.2) $12x^2 - 11x + 2$

Eg.3) $24x^2 - 10xy - y^2$

Eg.4.) $8a^2 + 13ab - 6b^2$

Homework: Ex. 10 pg. 28 (Mind Action Series new edition, 2022)

Homework: Ex. 11, questions B and C pg. 31, (Mind Action Series new edition, 2022)

Homework: Ex. 12 pg. 31 (Mind Action Series new edition, 2022)